

Answer Sheet for Module I part I Test I

Name An intermediate filament

April 7, 20

ID number _____

1 E		16 B		31 A	
2 B		17 A		32 B	
3 D		18 D		33 D	
4 C		19 B		34 C	
5 B		20 C		35 E	
6 E		21 A		36 B	
7 C		22 D		37 A	
8 D		23 B		38 B	
9 A		24 C		39 D	
10 C		25 C		40 D	
11 E		26 E		41 A	
12 D		27 C		42 B	
13 B		28 B		43 D	
14 C		29 D		44 D	
15 D		30 E		45 A	

INSTRUCTION 1: READ THE FOLLOWING QUESTIONS AND CHOOSE THE BEST ANSWER AND WRITE THE LETTER OF YOUR CHOICE ON THE SEPARATE ANSWER SHEET PROVIDED.

1. An intermediate filament present in the nuclear lamina and used for organization of the nuclear membrane and perinuclear chromatin is:
A. Vimentin
B. Desmin
C. Glial fibrillary acid protein
D. Cytokeratin
E. Lamin
2. Organelles involved in the lipid synthesis and detoxification are:
A. Rough endoplasmic reticulum
B. Smooth endoplasmic reticulum
C. Peroxisomes
D. Golgi bodies
E. Microtubules
3. Organelles made by stalk of 3-10 discrete flattened and slightly curved bag-like channels or cisternae are:
A. Mitochondria
B. Smooth endoplasmic reticulum
C. Peroxisomes
D. Golgi bodies
E. Microtubules
4. Organelles that contain many oxidative enzymes that produce hydrogen peroxide (H_2O_2) to kill microorganisms and detoxifies toxic agent are:
A. Rough endoplasmic reticulum
B. Smooth endoplasmic reticulum
C. Peroxisomes
D. Golgi bodies
E. Lysosomes
5. Which of the following are **not** inclusion?
A. Lipid droplets
B. Endosomes
C. Glycogen granules
D. Lipofuscin
E. Melanin granules
6. Which of the following is **not** correct regarding to the nucleolus:
A. Disappears during the cell division.
B. Is the site of ribosomal RNA synthesis.
C. Is nonmembrane bound.
D. Is made by regions of chromatins contributed by different chromosomes.
E. Is not visible under the light microscope.
7. The inactive X chromosome in the nucleus of epithelial cells from females is indicated as:
A. Nucleoplasm
B. Nucleoporins
C. Bar body
D. Nucleolar associated chromatin
E. Karyosomal chromatin

8. During a cell cycle,

- A. Duplication of centrioles begins and complete within the G_1 phase. ✓
- B. The S phase is the longest and most variable phase. ✓
- C. RNA and protein synthesis occur during the G_2 phase ✓
- D. Synthesis of DNA occurs during the S phase. ✓
- E. Replication of the chromosomes occurs during G_2 phase ✓

9. Who brought the study of anatomy as an objective discipline based on direct observations as well as scientific principles?

- A. Andreas Vesalius
- B. Hippocrates
- C. Aristotle

- D. Leonardo da Vinci
- E. William Harvey

10. An individual is performing the movement of flexion of the elbow joint when he/she:

- A. Moves the forearm away from the midline in the coronal plane.
- B. Moves the forearm posteriorly in the paramedian plane
- C. Moves the forearm anteriorly in the paramedian plane
- D. Rotates the forearm so that the anterior surface faces medially
- E. Moves the forearm toward the median sagittal plane.

11. Which one of the following is true about a patient who is standing in the anatomic position?

- A. Facing laterally
- B. Has the palms of the hands directed medially
- C. Has the ankles several inches apart ✓
- D. Is standing on his/her toes ✓
- E. Has the upper limbs by the sides of the trunk

12. The surgical notes of a patient state that she had a right infraumbilical paramedian incision through the skin of anterior abdominal wall. Where exactly was this incision anatomically?

- A. In the midline below the umbilicus ✓
- B. In the midline above the umbilicus ✓
- C. To the right of the midline above the umbilicus
- D. To the right of the midline below the umbilicus
- E. Just below the xiphoid process in the midline

13. A 20-year-old woman severely sprains her left ankle while playing tennis. When she tries to move the foot facing the sole medially, she experiences severe pain. What is the correct anatomic term for the movement of the foot that produces this pain?

- A. Pronation
- B. Inversion ✓
- C. Flexion

- D. Supination
- E. Eversion

14. Which criteria is used to classify bones as lamellar and non-lamellar bones?

- A. Location of bones
- B. Development of bones
- C. Microscopic appearance of bones

- D. Macroscopic appearance of bones
- E. Shape of bones

- the type of fracture where the bone is splintered into small fragments is termed as:
- A. Greenstick fracture
 - B. Impacted fracture
 - C. Compound fracture
 - D. Comminuted fracture
 - E. Fissured fracture
16. A type of fracture of either/both of distal ends of tibia & fibula at malleoli level is termed as:
- A. Colle's fracture
 - B. Pott's fracture
 - C. Avulsion
 - D. Transverse fracture
 - E. None of the above
17. Which type of joint is found between acromial end of scapula and clavicle?
- A. Plane joint
 - B. Hinge joint
 - C. Condylod joint
 - D. Pivot joint
 - E. Saddle joint
18. One of the following joint is typical example of saddle joint in our body:
- A. Metacarpophalangeal joint of index finger
 - B. Shoulder joint
 - C. Wrist joint
 - D. Carpometacarpal joint of the thumb
 - E. None of the above
19. Which one of the following joint is uniaxial type of joint? *plane, Hing*
- A. Saddle joint
 - B. Pivot joint
 - C. Condylod joint
 - D. Ball and socket joint
 - E. None of the above
20. Which one of the following elevation is not rounded elevation
- A. Tuberosity
 - B. Protuberance
 - C. Spinous process
 - D. Malleolus
 - E. None of the above
21. Which one of the following bone is not part of the axial skeleton of the human body?
- A. Clavicle
 - B. Ribs
 - C. Sternum
 - D. Cervical vertebral bones
 - E. Hyoid bone
22. A muscle that prevents unwanted movements in an intermediate joint so that another muscle can cross that joint and act primarily on a distal joint is termed as:
- A. Prime mover
 - B. Rotator
 - C. Fixator
 - D. Synergist
 - E. Antagonist
23. A characterstics of a muscle where it regains normal shape and size regardless of the degree of contraction is described by:
- A. Contractility
 - B. Elasticity
 - C. Extensibility
 - D. Irritability
 - E. Excitability

4. Which one of the following statement is not true about skeletal muscles?

- A. Its distal attachment is mostly fixed ✓
- B. It can have more than two attachments
- C. It has a role of musculovenous pump together with deep fascia
- D. Most externally covered by epimysium
- E. None of the above

25. Type of muscle contraction where it gives the muscle a certain firmness, assisting the stability of joints & the maintenance of posture is:

- A. Reflexive Contraction
- B. Isotonic contractions
- C. Tonic Contraction ✓
- D. Phasic Contraction
- E. Concentric contraction

26. Which criteria is not used in the naming of internal oblique abdominis and gluteus maximus muscles?

- A. Relative position of the muscle ✓
- B. Fiber arrangement of the muscle
- C. Location of the muscle ✓
- D. Relative size of the muscle ✓
- E. None of the above ✓

27. All of the following statements are true about body cavities and membranes except:

- A. Anterior body cavity is wider than posterior body cavity ✓
- B. Peritoneum is typical example of serous membrane that secrete watery fluid ✓
- C. Closed body cavities are lined by mucous membrane ✓
- D. Cutaneous membrane is the toughest membrane because of its location ✓
- E. Membranes are not always moist in the human body

28. Which one is not the function of nervous system in the human body?

- A. Control and coordinate the function of each organs ✓
- B. Create movement by contraction because of neurofilaments within it ✓
- C. Process and conduct nerve impulses ✓
- D. Initiate and control movements ✓
- E. None of the above

29. Which one does not describe collection of cell bodies of neurons in the nervous system:

- A. Nucleus in the central nervous system ✓
- B. Ganglia in the peripheral nervous system ✓
- C. Gray matter of the brain ✓
- D. White matter of the brain ✓
- E. Gray matter of the spinal cord

30. Which one of the following statement is not true about the nervous system?

- A. All spinal nerves are mixed which carry both motor and sensory fibers ✓
- B. Pain, temperature, touch & pressure are carried by general somatic afferent fibers ✓
- C. General visceral efferent fibers mainly constitute autonomic nervous system ✓
- D. Only three cranial nerves (CN-I, CN-II & CN-VIII) are purely sensory ✓
- E. None of the above ✓

31. All of the following cranial nerves are purely motor except:

- 3, 4, 6, 11, 12

- A. Trigeminal nerve
- B. Oculomotor nerve
- C. Trochlear nerve

- D. Hypoglossal nerve
- E. Accessory nerve

32. Which one of the following statement is not true about the circulatory system?

- A. All arteries carry blood away from the heart ✓
- B. Arteries that accompany deep veins are termed vena comitantes ✓
- C. Exchange of substances between extracellular fluid and blood occurs in the capillary bed ✓
- D. Veins always return blood towards the heart ✓
- E. None of the above

33. Vasodilation and vasoconstriction to regulate the blood that passes to the specific organs is mainly takes place at which level of the blood vessels?

- A. Elastic arteries
- B. Muscular arteries
- C. Conducting arteries (Aorta)
- D. Arterioles
- E. Capillaries

34. Which one is true about terminal arteries in the human body?

- A. True ^{preterminal} terminal arteries mainly supply segments of brain, liver, and kidneys ✓
- B. Arteries with ineffectual anastomoses forms anatomical end arteries ✓
- C. Occlusion of an end artery interrupts the blood supply to the segment of an organ it supplies ✓
- D. A and C are correct
- E. All of the above

35. All statements concerning vasculature is true except:

- A. Movement of blood through veins against gravity is assisted by contraction of skeletal muscles ✓
- B. Veins have more valves than arteries in the human body ✓
- C. Anastomoses can occur directly from arterioles to venules in some regions of the body ✓
- D. In portal venous system, blood passes through double capillary beds before reaching the heart ✓
- E. None of the above

36. Which one of the following statement is true concerning the lymphatic system?

- A. Lymphatic plexuses form anastomoses with the venules in the capillary bed ✓
- B. Lymphatic vessels are two groups as superficial and deep accompanying veins
- C. Lymphatic capillaries are found in all parts of the body where there is capillary beds ✓
- D. All lymphoid organs are involved in the filtration of lymph ✓
- E. All except C

37. Lymph from all of the following organs is collected to the thoracic duct except:

- A. Right lung
- B. Right kidney
- C. Left kidney

- D. Liver and gall bladder
- E. None of the above

38. Which one of the following lymphatic structures drains lymph directly from the tissues?
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- A. Lymphoid tissue ✓
- B. Lymph capillary
- C. Thoracic duct
- D. Right lymphatic duct
- E. Lymph node ✓

39. With regard to the preparation of tissues for microscopic examination,

- A. Paraffin is an embedding medium. ✓
- B. Clearing agents replace dehydrating agents by embedding agents.
- C. Infiltration is the introduction of embedding agents into the tissue. ✓
- D. All are correct
- E. None of the above

40. Following Haematoxylin & Eosin (H & E) staining, the basophilic components of the tissue:

- A. Stain dark blue ✓
- B. Include the nucleus
- C. May show metachromatic reaction
- D. All of the above
- E. None of the above

41. Which of the following is the function of ocular lens of light microscope?

- A. Magnification
- B. Resolution
- C. Focusing light to the specimen
- D. A & B
- E. All of the above

42. The plasma membrane:

- A. Has peripheral proteins which are mostly found loosely attached on the outer surface of the membrane. ✓
- B. Has integral proteins which show free lateral mobility and rearrangements. ✓
- C. Has oligosaccharides which are found projecting from both outer and inner surfaces. ✓
- D. Is clearly visible under light microscopy
- E. All are correct

43. Which of the following is not a membranous organelle?

- A. Lysosomes ✓
- B. Peroxisomes ✓
- C. Mitochondria ✓
- D. Ribosomes
- E. Endoplasmic reticulum

44. Centrioles are constructed from:

- A. Lysosomes
- B. Peroxisomes
- C. Microfilaments
- D. Microtubules
- E. Intermediate filaments

45. Which of the following organelles are used for intracellular transport system?

- A. Microtubules
- B. Peroxisomes ✓
- C. Microfilaments
- D. Smooth endoplasmic reticulum.
- E. Intermediate filaments